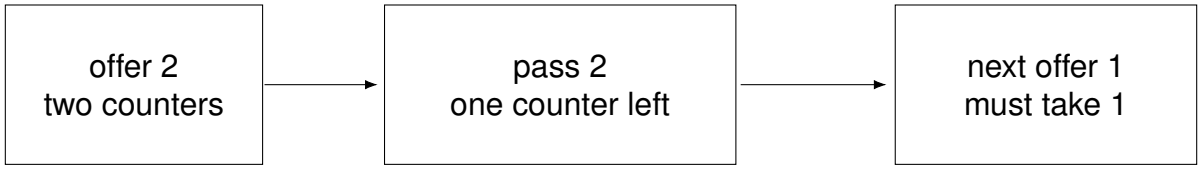
Each bag has one ticket marked 0, one marked 4 and one marked 6. Each has the same chance. Every offer is a fresh draw from the whole bag: roll the bag's die, or put the ticket back and mix.

Set the number of offers before a round, and put out one turn counter for each offer. Your partner shows each offer. Take it to end the round, or pass it forever. With one counter left, you must take the offer, even if it is 0. Remove one turn counter after deciding on each offer. Only the offer you take scores.

A practice round uses different tickets: 1, 2, 5.



round score: 1

Problem 1: Use the 0, 4, 6 bag. Invent a taking rule for two-offer rounds and one for three-offer rounds. Keep each rule fixed while playing six rounds with a partner. Try to get a high total score. Swap roles.

Two-offer rule _____

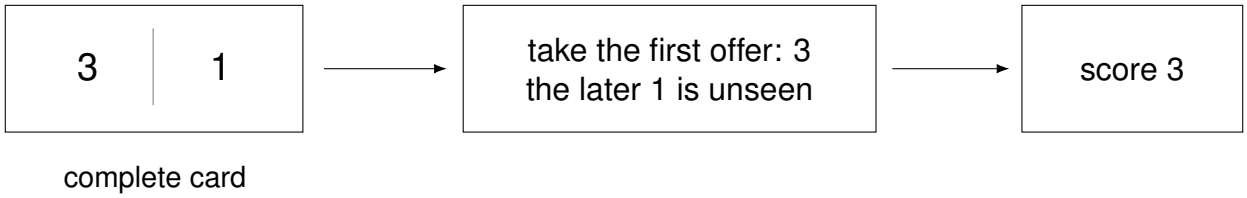
Three-offer rule _____

Round	1	2	3	4	5	6	Total
Two offers							
Three offers							

Problem 2: Can your fixed two-offer rule beat your three-offer rule in one six-round match and lose to it in another? Show possible offers for both matches. Can either match settle which rule has the better average over many rounds?

A card lists offers in drawing order, including offers that would remain unseen after you take one. All complete cards of the same length have the same chance.

With practice tickets 1, 3, 5, this rule takes any first offer:



Problem 3: Find first-offer choices for 0, 4 and 6 that give the highest total score on all nine cards. Use the same choices on every card. The second offer must be taken if you reach it.

0 | 0 score _____	0 | 4 score _____	0 | 6 score _____
4 | 0 score _____	4 | 4 score _____	4 | 6 score _____
6 | 0 score _____	6 | 4 score _____	6 | 6 score _____

First offer 0: _____

First offer 4: _____

First offer 6: _____

Total score on nine cards: _____

Problem 4: Which plan gives the higher total score on these 27 three-offer cards? Use every card for each plan, including cards with the same score.

A: Take 4 or 6 whenever it appears. Pass 0.

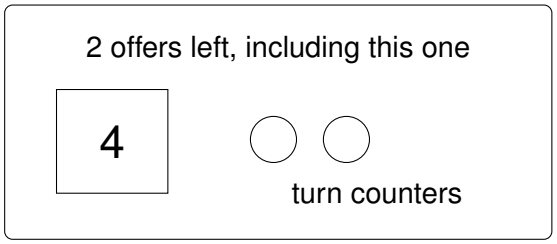
B: Take only 6 on the first offer. On the second, take 4 or 6 and pass 0.

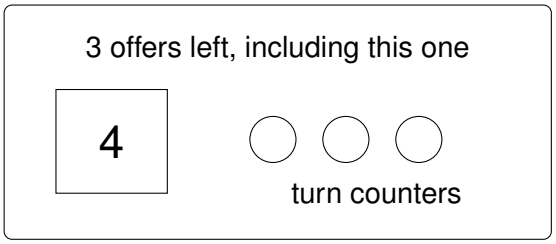
Both plans take the last offer if they reach it.

A's total: _____ B's total: _____

0 | 0 | 0 A _____ B _____	0 | 0 | 4 A _____ B _____	0 | 0 | 6 A _____ B _____
0 | 4 | 0 A _____ B _____	0 | 4 | 4 A _____ B _____	0 | 4 | 6 A _____ B _____
0 | 6 | 0 A _____ B _____	0 | 6 | 4 A _____ B _____	0 | 6 | 6 A _____ B _____
4 | 0 | 0 A _____ B _____	4 | 0 | 4 A _____ B _____	4 | 0 | 6 A _____ B _____
4 | 4 | 0 A _____ B _____	4 | 4 | 4 A _____ B _____	4 | 4 | 6 A _____ B _____
4 | 6 | 0 A _____ B _____	4 | 6 | 4 A _____ B _____	4 | 6 | 6 A _____ B _____
6 | 0 | 0 A _____ B _____	6 | 0 | 4 A _____ B _____	6 | 0 | 6 A _____ B _____
6 | 4 | 0 A _____ B _____	6 | 4 | 4 A _____ B _____	6 | 4 | 6 A _____ B _____
6 | 6 | 0 A _____ B _____	6 | 6 | 4 A _____ B _____	6 | 6 | 6 A _____ B _____

Problem 5: The current offer is 4. Take or pass in each position? Explain with average scores.





Choice: _____

Choice: _____

Problem 6: Can any legal plan beat plan B from Problem 4 on the 27 cards? Explain. A plan is allowed to remember earlier passed offers.

Problem 7: For each new bag, find all the first-offer choices that give the highest total score on its nine two-offer cards.

Bag: 0, 3, 6

0 | 0 score _____	0 | 3 score _____	0 | 6 score _____
3 | 0 score _____	3 | 3 score _____	3 | 6 score _____
6 | 0 score _____	6 | 3 score _____	6 | 6 score _____

Choices: _____ Total: _____

Bag: 0, 5, 6

0 | 0 score _____	0 | 5 score _____	0 | 6 score _____
5 | 0 score _____	5 | 5 score _____	5 | 6 score _____
6 | 0 score _____	6 | 5 score _____	6 | 6 score _____

Choices: _____ Total: _____

Problem 7 (continued)

Bag: 0, 2, 6

0 | 0 score _____	0 | 2 score _____	0 | 6 score _____
2 | 0 score _____	2 | 2 score _____	2 | 6 score _____
6 | 0 score _____	6 | 2 score _____	6 | 6 score _____

Choices: _____ Total: _____

Problem 8: Use a bag with tickets 0, 5, 6. Find the best taking plan for three-offer rounds and for four-offer rounds. Give each plan's average score.

Offers	Taking plan	Average
Three		
Four		

Problem 9: The bag and draw rules stay fixed. Could adding one allowed offer lower the best possible average score? Could it leave that average unchanged? Explain your answers for any known bag of score tickets.